

```
In [85]: from sklearn import preprocessing
import pandas as pd
import numpy as np
from sklearn.model_selection import train_test_split
from sklearn.naive_bayes import GaussianNB
import matplotlib.pyplot as plt
from sklearn.metrics import confusion_matrix
from sklearn.metrics import accuracy_score, precision_score, recall_score, f1_score
```

```
In [86]: df=pd.read_csv("IRIS.csv")
```

```
In [87]: df
```

```
Out[87]:
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	Id	SepalLengthCm	SepalWidthCm	PetalLengthCm	PetalWidthCm	Species
0	1	5.1	3.5	1.4	0.2	Iris-setosa
1	2	4.9	3.0	1.4	0.2	Iris-setosa
2	3	4.7	3.2	1.3	0.2	Iris-setosa
3	4	4.6	3.1	1.5	0.2	Iris-setosa
4	5	5.0	3.6	1.4	0.2	Iris-setosa
...	...	...	...	...	...	...
145	146	6.7	3.0	5.2	2.3	Iris-virginica
146	147	6.3	2.5	5.0	1.9	Iris-virginica
147	148	6.5	3.0	5.2	2.0	Iris-virginica
148	149	6.2	3.4	5.4	2.3	Iris-virginica
149	150	5.9	3.0	5.1	1.8	Iris-virginica

150 rows × 6 columns

```
In [88]: df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 150 entries, 0 to 149
Data columns (total 6 columns):
#   Column          Non-Null Count  Dtype
---  -
0   Id               150 non-null   int64
1   SepalLengthCm    150 non-null   float64
2   SepalWidthCm     150 non-null   float64
3   PetalLengthCm    150 non-null   float64
4   PetalWidthCm     150 non-null   float64
5   Species          150 non-null   object
dtypes: float64(4), int64(1), object(1)
memory usage: 7.2+ KB
```

```
In [89]: label_enc=preprocessing.LabelEncoder()
```

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In [90]: df['Species']=label_enc.fit_transform(df['Species'])
```

```
In [91]: df
```

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Out[91]:
```

	Id	SepalLengthCm	SepalWidthCm	PetalLengthCm	PetalWidthCm	Species
0	1	5.1	3.5	1.4	0.2	0
1	2	4.9	3.0	1.4	0.2	0
2	3	4.7	3.2	1.3	0.2	0
3	4	4.6	3.1	1.5	0.2	0
4	5	5.0	3.6	1.4	0.2	0
...	...	...	...	...	...	...
145	146	6.7	3.0	5.2	2.3	2
146	147	6.3	2.5	5.0	1.9	2
147	148	6.5	3.0	5.2	2.0	2
148	149	6.2	3.4	5.4	2.3	2
149	150	5.9	3.0	5.1	1.8	2

150 rows × 6 columns

```
In [92]: #independent variable
x=df.drop("Species",axis=1)

#dependent varibale
y=df["Species"]
```

```
In [93]: print(x.head())
```

	Id	SepalLengthCm	SepalWidthCm	PetalLengthCm	PetalWidthCm
0	1	5.1	3.5	1.4	0.2
1	2	4.9	3.0	1.4	0.2
2	3	4.7	3.2	1.3	0.2
3	4	4.6	3.1	1.5	0.2
4	5	5.0	3.6	1.4	0.2

```
In [94]: print(y.head())
```

```
0    0
1    0
2    0
3    0
4    0
Name: Species, dtype: int64
```

```
In [95]: x_train, x_test, y_train, y_test = train_test_split(x,y, test_size=0.2)
        model=GaussianNB()
        model.fit(x_train,y_train)
        y_pred=model.predict(x_test)
```

```
In [96]: print("Predictions:", y_pred)
```

Predictions: [2 2 0 2 1 1 1 1 2 1 2 2 0 2 0 0 2 2 1 2 2 1 0 0 1 1 0 0 1 2]

```
In [97]: cm= confusion_matrix(y_test,y_pred)
        cm
```

```
Out[97]: array([[ 8,  0,  0],
               [ 0, 10,  0],
               [ 0,  0, 12]])
```

```
In [98]: print(cm.ravel())
```

[8 0 0 0 10 0 0 0 12]

```
In [99]: cm_array=np.array(cm)
```

```
In [100... cm_array
```

```
Out[100... array([[ 8,  0,  0],
               [ 0, 10,  0],
               [ 0,  0, 12]])
```

```
In [101... total = cm_array.sum()
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```
In [102... total
```

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Out[102... np.int64(30)
```

```
In [103... for i in range(len(cm)):
            TP = cm[i, i]
            FP = cm[:, i].sum() - TP
            FN = cm[i, :].sum() - TP
            TN = total - (TP + FP + FN)
            print(f"Class {i}: TP={TP}, FP={FP}, FN={FN}, TN={TN}")
```

Class 0: TP=8, FP=0, FN=0, TN=22
 Class 1: TP=10, FP=0, FN=0, TN=20
 Class 2: TP=12, FP=0, FN=0, TN=18

```
In [105... precision= precision_score(y_test, y_pred,average='micro')
        print("precision= ",precision)
```

precision= 1.0

```
In [106... Accuracy= accuracy_score(y_test, y_pred)
        print("accuracy= ",Accuracy)
```

accuracy= 1.0

```
In [108... F1score= f1_score(y_test, y_pred,average='micro')  
print("F1score= ",F1score)
```

```
F1score= 1.0
```

```
In [110... Recallscore= recall_score(y_test, y_pred,average='micro')  
print("recall= ",Recallscore)
```

```
recall= 1.0
```